

# Session Zero Build Guide

Tools, responsibilities, and how to wire the Web3 game demo without constructing a doomed mini-studio.

## Project shape

Build a browser-based, mage-only turn-based combat game. The demo proves one UX claim: a player can sign in with Google, approve a single limited match session, play without wallet interruptions, buy a consumable inside combat, and inspect labelled Arbitrum testnet activity afterward.

## The stack in one sentence

**SvelteKit** owns the app shell and Web3 UI. **Phaser** owns the battle canvas. **Magic** owns Google login and embedded wallet creation. **ZeroDev** owns smart-account execution, session permissions, paymaster gas sponsorship and batching. **Arbitrum Sepolia** proves real onchain settlement.

| Tool                | What it does                                       | How you use it in Session Zero                                                                                          |
|---------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| SvelteKit           | Frontend app framework.                            | Routes, landing page, login screen, session-spend screen, game shell, post-match report.                                |
| Phaser 3            | 2D game engine for browser canvas.                 | Combat scene, sprites, spell effects, enemy attack timing, shield input windows.                                        |
| Magic               | Embedded wallet + social login provider.           | Google login creates an embedded EOA. Optional production fiat on-ramp via wallet widget.                               |
| ZeroDev             | Smart accounts, paymasters, permissions, batching. | Use the Magic signer to control a Kernel smart account. Create one scoped match session. Sponsor gas. Batch settlement. |
| Arbitrum Sepolia    | Arbitrum testnet.                                  | Deploy contracts and show real transaction hashes in the demo.                                                          |
| Solidity + Hardhat  | Smart contract language + JS tooling.              | Deploy GameToken and SessionZeroGame contracts.                                                                         |
| viem                | Type-safe EVM client library.                      | Read contracts, send user operations through ZeroDev flows, format tx links.                                            |
| Tailwind CSS        | Utility CSS.                                       | Fast UI styling for the non-game screens.                                                                               |
| Arbiscan/Blockscout | Explorers.                                         | Proof links for start match, purchase, and match settlement.                                                            |

# Recommended architecture

Keep the player-facing flow tiny. Hide the complicated controls under defaults that your code enforces. Humans do not queue for games so they can configure function selectors. Civilization has fallen far enough.

## Runtime flow

```
Google login -> Magic EOA -> ZeroDev Kernel account -> one match session permission -> Phaser battle -> real Arbitrum Sepolia transactions -> labelled report.
```

## Important account note

The standard Magic + ZeroDev guide uses the Magic account as signer/owner for a ZeroDev smart account. That can create two technical addresses, but it should not create two user onboarding flows. Display only the player identity and game account, not a wallet anatomy lesson.

## Same-address future path

Magic and ZeroDev both document EIP-7702 paths. The clean production direction is one embedded EOA upgraded with smart-account capabilities such as gas sponsorship, batching and session keys. Treat that as a stretch/future item unless you prove it end to end before the demo.

# Build order

- **1. Build the battle with mocked Web3:** HP, mana, attacks, shields, potions, victory summary.
- **2. Add contracts:** mock gUSDC faucet and one SessionZeroGame contract with startMatch, buyAndUsePotion, completeMatch.
- **3. Add Magic:** Google login, get the embedded wallet signer/provider.
- **4. Add ZeroDev:** create Kernel account, sponsor one simple transaction first.
- **5. Add session permissions:** one match limit, allowed game contract, allowed purchase function, expiry.
- **6. Replace mocked purchase:** Buy-and-use calls the real contract, but gameplay continues optimistically.
- **7. Add settlement batch:** complete match, record action digest, mint reward/receipt, close session.
- **8. Add explorer summary:** human labels first, hashes and links second.

# Session permission design

The player chooses only the maximum spend. The application silently supplies the boring safety rules.

## Player sees:

Maximum spend this match: 1.00 gUSDC Valid until: this match ends Gas fees: sponsored  
Allowed: approved consumables only Not allowed: withdrawals, transfers, arbitrary contract calls

## System enforces:

- Allowed contract: SessionZeroGame only
- Allowed function: buyAndUsePotion plus match settlement functions
- Allowed token: gUSDC only
- Spend cap: chosen by player
- Expiry: match end or fallback timer
- No transferFrom to random recipients
- No NFT transfers or withdrawals

# Game system guide

Use mage-only combat. Magic shields and projectiles hide the fact that your characters do not have complex physical animation. This is called scope control, not cowardice, despite what the feature goblin in your brain says.

| Element           | Implementation                                                              |
|-------------------|-----------------------------------------------------------------------------|
| Renderer          | Phaser canvas mounted inside a Svelte route.                                |
| Characters        | 2.5D transparent PNG/WebP poses: idle, cast, hit, defend, defeated/victory. |
| Battle view       | Three-quarter side view with player bottom-right and enemy upper-left.      |
| Attacks           | Small sparks, medium lance, large area burst.                               |
| Shields           | Small/medium/large magic wards with short active windows.                   |
| Resources         | Mana powers attacks and shields. Health and mana potions are consumables.   |
| Blockchain action | Purchasing/using a potion and match settlement, not every attack.           |

## Combat model

Small attack costs 1 mana and fires three fast sparks. Medium attack costs 2 mana and fires one larger blast. Large attack costs 4 mana and creates an area explosion. Small shield costs 1 mana and blocks small attacks if timed correctly. Medium shield costs 2 mana and blocks small/medium. Large shield costs 4 mana and blocks all. Large shields waste mana against bait attacks, creating the strategy loop.

## Do not put every attack onchain

Attacks, blocks, hit calculations and animations happen offchain for speed. Use onchain operations for match start, consumable purchase/use, final result, rewards and an action-log digest. This is the sane hybrid model. Putting every fireball onchain is how a game becomes a distributed slideshow.

# Minimal folder plan

```
src/lib/web3/magic.js src/lib/web3/zerodev.js src/lib/web3/contracts.js
src/lib/game/SessionZeroScene.js src/lib/game/combatState.js src/lib/game/effects.js
src/lib/stores/playerSession.js src/routes/+page.svelte src/routes/play/+page.svelte
src/routes/report/+page.svelte contracts/GameToken.sol contracts/SessionZeroGame.sol
scripts/deploy.js
```

## Development checklist

- Deployed frontend URL works without local environment variables leaking.
- Google login works in a fresh browser profile.
- New player gets test gUSDC automatically or by one visible faucet button.
- One session authorization is required before battle.
- Potion purchase generates a real Arbitrum Sepolia transaction.
- No wallet prompt appears during battle after the session approval.
- Match summary has at least three labelled onchain entries.
- Explorer links open correctly.
- Demo video shows the problem before mechanisms.

## Sources and docs

|                                |                                                                                                                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BGA 2024 onboarding UX         | <a href="https://blockchaingamealliance.net/user-acquisition-in-web3-gaming-xspaces/">https://blockchaingamealliance.net/user-acquisition-in-web3-gaming-xspaces/</a>                         |
| DappRadar Q2 2025 Web3 Gaming  | <a href="https://dappradar.com/blog/state-of-blockchain-gaming-in-q2-2025">https://dappradar.com/blog/state-of-blockchain-gaming-in-q2-2025</a>                                               |
| ZeroDev Magic signer           | <a href="https://docs.zerodev.app/sdk/v5_3_x/signers/magic">https://docs.zerodev.app/sdk/v5_3_x/signers/magic</a>                                                                             |
| ZeroDev session keys           | <a href="https://docs.zerodev.app/sdk/v5_3_x/advanced/session-keys">https://docs.zerodev.app/sdk/v5_3_x/advanced/session-keys</a>                                                             |
| ZeroDev permissions            | <a href="https://docs.zerodev.app/smart-accounts/permissions/intro">https://docs.zerodev.app/smart-accounts/permissions/intro</a>                                                             |
| ZeroDev transaction automation | <a href="https://docs.zerodev.app/sdk/v5_3_x/permissions/transaction-automation">https://docs.zerodev.app/sdk/v5_3_x/permissions/transaction-automation</a>                                   |
| Magic wallet widget/onramp     | <a href="https://docs.magic.link/embedded-wallets/wallets/customization/widget-ui">https://docs.magic.link/embedded-wallets/wallets/customization/widget-ui</a>                               |
| Magic signatures               | <a href="https://docs.magic.link/embedded-wallets/wallets/features/personal-signatures">https://docs.magic.link/embedded-wallets/wallets/features/personal-signatures</a>                     |
| Magic EIP-7702                 | <a href="https://docs.magic.link/embedded-wallets/wallets/features/eip-7702">https://docs.magic.link/embedded-wallets/wallets/features/eip-7702</a>                                           |
| Arbitrum Sepolia quickstart    | <a href="https://docs.arbitrum.io/build-decentralized-apps/quickstart-solidity-remix">https://docs.arbitrum.io/build-decentralized-apps/quickstart-solidity-remix</a>                         |
| Arbitrum explorers             | <a href="https://docs.arbitrum.io/arbitrum-essentials/reference/monitoring-tools-block-explorers">https://docs.arbitrum.io/arbitrum-essentials/reference/monitoring-tools-block-explorers</a> |
| Phaser animations              | <a href="https://docs.phaser.io/phaser/concepts/animations">https://docs.phaser.io/phaser/concepts/animations</a>                                                                             |
| SvelteKit docs                 | <a href="https://svelte.dev/docs/kit/introduction">https://svelte.dev/docs/kit/introduction</a>                                                                                               |
| Hardhat docs                   | <a href="https://hardhat.org/docs">https://hardhat.org/docs</a>                                                                                                                               |
| Viem docs                      | <a href="https://viem.sh/docs/getting-started">https://viem.sh/docs/getting-started</a>                                                                                                       |
| EIP-712                        | <a href="https://eips.ethereum.org/EIPS/eip-712">https://eips.ethereum.org/EIPS/eip-712</a>                                                                                                   |